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Experiment 8

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Branch: BE CSE **Section/Group: KRG 3A**
Semester: 6th **Date of Performance: 06/04/26**
Subject Name: Full Stack Development **Subject Code: 23CSH-309**

Aim:
To understand and implement security in a full-stack application (LivePoll System) using Spring Security, OAuth (Google Login), role-based access control (RBAC), and integration with a React frontend.

Objective:
1. To understand the importance of security in backend applications
2. To learn how Spring Security manages authentication and authorization
3. To differentiate between authentication and authorization
4. To secure REST APIs using custom security configurations
5. To implement OAuth 2.0 (Google Login)
6. To apply role-based access control (USER, ADMIN)

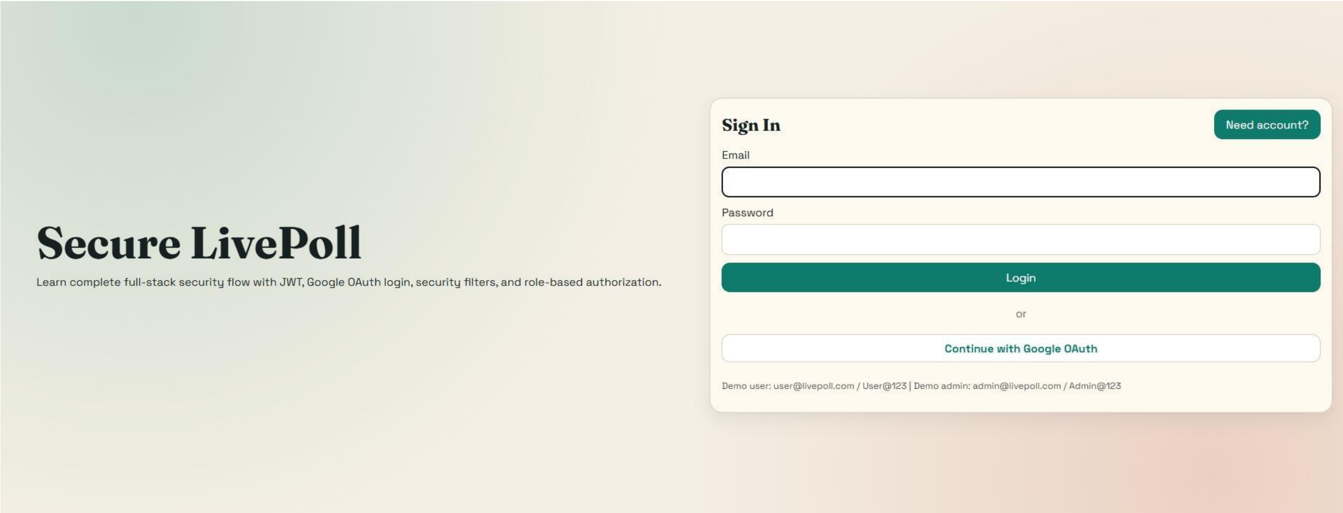
Implementation/Code:

AdminPollController.java

```
package com.ecotrack.livepoll.controller;

import com.ecotrack.livepoll.auth.AppUserPrincipal;
import com.ecotrack.livepoll.dto.PollDtos;
import com.ecotrack.livepoll.service.PollService;
import jakarta.validation.Valid;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.security.core.annotation.AuthenticationPrincipal;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
```

```
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
@RequestMapping("/api/admin/polls")
@PreAuthorize("hasRole('ADMIN')")
public class AdminPollController {

    private final PollService pollService;

    public AdminPollController(PollService pollService) {
        this.pollService = pollService;
    }

    @PostMapping
    public ResponseEntity<PollDtos.PollResponse> createPoll(@Valid @RequestBody
PollDtos.CreatePollRequest request,
                                                            @AuthenticationPrincipal AppUserPrincipal
principal) {
        return ResponseEntity.status(HttpStatus.CREATED)
                .body(pollService.createPoll(request, principal.getUsername()));
    }

    @PostMapping("/{id}/close")
    public ResponseEntity<PollDtos.PollResponse> closePoll(@PathVariable Long id) {
        return ResponseEntity.ok(pollService.closePoll(id));
    }

    @DeleteMapping("/{id}")
    public ResponseEntity<Void> deletePoll(@PathVariable Long id) {
        pollService.deletePoll(id);
        return ResponseEntity.noContent().build();
    }
}
```

Output:



Learning Outcome:

- Learned how Spring Security secures backend APIs
- Understood authentication vs authorization clearly
- Implemented OAuth 2.0 login using Google
- Gained experience with role-based access control (RBAC)
- Learned how to integrate React frontend with secured backend
- Understood complete request flow with security filters
- Improved debugging and analysis of secured applications